MICROSOFT AZURE

Cloud Governance & Resource Organization

Technical Documentation Report

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1. Project Overview

This document presents the complete technical implementation of Azure Cloud Governance and Resource Organization, demonstrating foundational principles of cloud resource management within Microsoft Azure. The project covers subscription management, resource group organization, tagging strategies, and Role-Based Access Control (RBAC) implementation.

Project Attribute	Details
Cloud Platform	Microsoft Azure
Subscription	Azure Subscription 1
Region	West Europe (Primary)
Resource Groups	3 (dev-webapp-westeurope-rg, prod-webapp-westeurope-rg, temp-test-westeurope-rg)
IAM Framework	Azure Role-Based Access Control (RBAC)
Tagging Strategy	Environment-Team key-value metadata tags

2. Naming Convention Strategy

A consistent, descriptive naming convention was adopted across all Azure resources to support clarity, filtering, cost tracking, and automation. The naming format follows a structured pattern that encodes the workload purpose, environment tier, target region, and resource type.

2.1 Naming Pattern

```
Format:   {workload}-{environment}-{region}-{resource-type}

Example:  dev-webapp-westeurope-rg
```

2.2 Naming Component Reference

Component	Values Used	Purpose
Workload	webapp, test	Identifies the application or workload type
Environment	dev, prod, temp-test	Differentiates deployment environment tier
Region	westeurope	Azure region where resources are deployed
Resource Type	rg, vnet, st01, sql01	Abbreviation of the Azure resource type

2.3 Resource Naming Examples

Resource Name	Type	Environment
dev-webapp-westeurope-rg	Resource Group	Development
prod-webapp-westeurope-rg	Resource Group	Production
temp-test-westeurope-rg	Resource Group	Temporary / Test
dev-webapp-westeurope-vnet	Virtual Network	Development
prodwebappwesteuropest01	Storage Account	Production
temptestwesteuropesvr01	SQL Server	Temporary / Test
temptestwesteuropesql01	SQL Database	Temporary / Test

3. Resource Group Architecture

Three resource groups were provisioned under Azure Subscription 1 to represent distinct operational environments. Each group serves as an isolated lifecycle boundary, enabling independent policy application, cost tracking, and access control.

3.1 Resource Group Summary

Resource Group	Environment	Region	Tag Applied
dev-webapp-westeurope-rg	Development	West Europe	3mttdevop: teamA
prod-webapp-westeurope-rg	Production	West Europe	3mttwebapp: teamB
temp-test-westeurope-rg	Temp / Test	South Africa North*	3mttwebapp: teamC

* Note: Resources within temp-test-westeurope-rg (SQL Server and SQL Database) were deployed to South Africa North, demonstrating that resources within a resource group can span different regions from the group itself.

3.2 Deployed Resources per Group

dev-webapp-westeurope-rg (Development)

Resource Name	Type	Location
dev-webapp-westeurope-vnet	Virtual Network	West Europe

prod-webapp-westeurope-rg (Production)

Resource Name	Type	Location
prodwebappwesteuropest01	Storage Account	West Europe

temp-test-westeurope-rg (Temporary / Test)

Resource Name	Type	Location
temptestwesteuropesql01 (temptestwesteuropesvr01)	SQL Database	South Africa North
temptestwesteuropesvr01	SQL Server	South Africa North

4. Tagging Strategy

Azure resource tags were applied as key-value metadata pairs to all resource groups. Tags serve as the primary mechanism for cost allocation, team accountability, and automated resource filtering. The tagging design follows a two-dimension model: project identifier and team ownership.

4.1 Tag Schema

Tag Key	Tag Value	Scope	Purpose
3mttdevop	teamA	dev-webapp-westeurope-rg	DevOps workload ownership
3mttwebapp	teamB	prod-webapp-westeurope-rg	Web App production ownership
3mttwebapp	teamC	temp-test-westeurope-rg	Web App test ownership

4.2 Tagging Rationale

The tagging strategy was designed around the following governance principles:

- Cost Allocation: Tags allow Azure Cost Management to filter and aggregate spend by team or project, enabling accurate chargeback and showback reporting.
- Team Accountability: Each resource group is explicitly owned by a named team, establishing clear accountability boundaries for resource provisioning and decommission.
- Automation Enablement: Tag-based filtering can trigger automated workflows (e.g., auto-shutdown dev resources at night) through Azure Policy or Azure Automation runbooks.
- Naming Consistency: Tag keys use the project prefix (3mtt) combined with workload type (devop / webapp), and values use simple team identifiers (teamA, teamB, teamC) that are easy to filter and reference in scripts.

5. Role-Based Access Control (RBAC)

RBAC was implemented at the Resource Group level to enforce the principle of least privilege across different team personas. Access levels were designed to simulate a real multi-team cloud environment where developers, operators, and administrators have differentiated permissions.

5.1 RBAC Assignments — prod-webapp-westeurope-rg

Principal	Type	Role	Scope	Condition
Nzegbuna Mic... (42a2472c)	User	Owner	Subscription (Inherited)	None
Nzegbuna Mic... (42a2472c)	User	Owner	Subscription (Inherited)	None
nzegbunachuk...	User	Reader	This resource	None

5.2 RBAC Assignments — dev-webapp-westeurope-rg

Principal	Type	Role	Scope	Cond.
Nzegbuna Mic... (42a2472c)	User	Owner	Subscription (Inherited)	None
Nzegbuna Mic... (42a2472c)	User	Owner	Subscription (Inherited)	None
nzegbunachuk...	User	App Configuration Contributor	This resource	None

5.3 RBAC Design Rationale

Role	Assigned To	Rationale
Owner	Primary Account	Full administrative control over the subscription and all child resource groups. Inherited from subscription scope to apply consistently across all environments.
Reader	nzegbunachuk (prod)	Read-only access granted at the Production resource group scope. Appropriate for audit roles, monitoring personnel, or stakeholders who need visibility without modification rights.
App Configuration Contributor	nzegbunachuk (dev)	Scoped contributor access at the Development resource group level. Enables developers to manage application configuration resources without broader infrastructure permissions — enforcing least privilege.

5.4 RBAC Scope Hierarchy

The RBAC assignments leverage Azure's scope inheritance model:

- Subscription Scope (Owner): Permissions inherited by all child resource groups automatically.
- Resource Group Scope (Reader / Contributor): Targeted, least-privilege assignments that do not propagate upward.
- This two-tier approach ensures administrative users maintain full control while restricting external or junior team members to only the resources relevant to their role.

6. Resource Lifecycle Management

6.1 Temporary Resource Group Deletion

The temp-test-westeurope-rg resource group was used to demonstrate bulk lifecycle management. After validating its SQL Server and SQL Database resources, the group's resources were deleted to simulate decommissioning a temporary testing environment.

Lifecycle Step	Description
Provision	SQL Server (temptestwesteuropesvr01) and SQL Database deployed to South Africa North region within the temp group.
Tag & Govern	Tag 3mttwebapp: teamC applied. RBAC not separately assigned as no persistent team access needed.
Validate	Resources verified as operational; confirmed database connectivity to server within the group.
Delete	Resources removed (screenshot shows empty resource group: 'No resources match your filters'). Group retained as empty container.

This exercise validated that deleting resources within a group does not delete the group itself, and demonstrated how resource groups serve as logical containers that persist independently of their child resources.

7. Governance Architecture Summary

The following diagram illustrates the complete governance hierarchy implemented across the Azure subscription:

Azure Subscription 1 (Billing & Policy Boundary)

Owner: michaelnzegbuna@gmail.com | RBAC Scope: Subscription-level

```
  +-- dev-webapp-westeurope-rg  [Tag: 3mttdevop=teamA]  [RBAC: App Config
Contributor]
```

```
  |     +-- dev-webapp-westeurope-vnet  (Virtual Network, West Europe)

  +-- prod-webapp-westeurope-rg  [Tag: 3mttwebapp=teamB]  [RBAC: Reader]
  |     +-- prodwebappwesteuropest01  (Storage Account, West Europe)

  +-- temp-test-westeurope-rg  [Tag: 3mttwebapp=teamC]  [Resources
deleted]
        +-- [DELETED] temptestwesteuropesvr01  (SQL Server)
        +-- [DELETED] temptestwesteuropesql01  (SQL Database)
```

8. Key Learnings & Observations

- Resource Group as Lifecycle Unit: Grouping resources by environment and workload enables bulk operations (tagging, deletion, policy) to be applied efficiently without impacting other environments.
- Naming Convention Value: A consistent naming schema made resource identification, filtering, and cost analysis immediately intuitive without relying on additional documentation.
- Scope-Based RBAC: Assigning permissions at the most specific required scope (resource group vs. subscription) reduces the blast radius of credential compromise and enforces least privilege.
- Tag Inheritance Limitation: Tags applied at the resource group level do not automatically propagate to child resources — a consideration for cost allocation completeness requiring Azure Policy or manual tag assignment.
- Regional Flexibility: Resources within a single resource group can span multiple Azure regions, offering architectural flexibility while maintaining governance boundaries.
- Temp Group Pattern: Using a dedicated temporary resource group for experimental workloads provides clean isolation and trivial decommissioning without affecting production or development environments.

Appendix: Azure CLI Reference Commands

The following Azure CLI commands can be used to replicate this resource organization structure programmatically:

```
# Create Resource Groups
az group create --name dev-webapp-westeurope-rg --location westeurope
az group create --name prod-webapp-westeurope-rg --location westeurope
az group create --name temp-test-westeurope-rg --location westeurope

# Apply Tags
az group update --name dev-webapp-westeurope-rg --tags 3mttdevop=teamA
az group update --name prod-webapp-westeurope-rg --tags 3mttwebapp=teamB
az group update --name temp-test-westeurope-rg --tags 3mttwebapp=teamC

# Assign RBAC Roles
az role assignment create --role Reader \
```

```
  --assignee <user-id> --scope /subscriptions/<sub-id>/resourceGroups/prod-webapp-westeurope-rg

# Delete temp resource group
az group delete --name temp-test-westeurope-rg --yes --no-wait
```

End of Document | Azure Cloud Governance & Resource Organization | 3MTT DevOps Track | May 2026